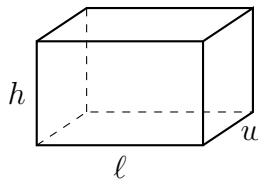


Surface Area, Composite Shapes, and Scale Design

Surface area of prisms, composite areas, and scale-drawing design

Directions:

- These problems mix four skills; after the warm-up, each problem makes you choose the method. Show every step and include the unit.
- The figure below shows how every box in this set is labelled — length ℓ , width w , height h . No values are shown; use the numbers given in each problem.



- 1. Warm-up.** A closed box is 4 cm long, 3 cm wide, and 2 cm tall. Find its total surface area.

Answer: _____ cm^2

- 2.** A cube-shaped tissue box has edges 5 inches long. Find its total surface area.

Answer: _____ in^2

3. A blueprint uses the scale 1 cm : 4 m. A wall measures 3.5 cm on the blueprint. How long is the real wall?

Answer: _____ m

4. On a meter grid, an L-shaped patio has corners at (0, 0), (8, 0), (8, 4), (5, 4), (5, 7), and (0, 7). Split it into rectangles and find its area.

Answer: _____ m²

5. A floor plan uses the scale 1 in : 5 ft. A bedroom measures 2 in by 3 in on the plan. Find the area of the *real* bedroom. (*Careful: scale each side first.*)

Answer: _____ ft²

6. A gift box is 6 in long, 4 in wide, and 3 in tall.

(a) How much wrapping paper is needed to cover all six faces exactly?

(b) Paper is sold in sheets of 60 square inches. How many whole sheets must you buy?

Answer: _____ in²

7. On a meter grid, a T-shaped stage has corners at (0, 0), (10, 0), (10, 3), (7, 3), (7, 8), (3, 8), (3, 3), and (0, 3). Find its area.

Answer: _____ m²

8. A model bridge is drawn at the scale 1 cm : 8 m. The real bridge is 120 m long. How long is the bridge on the drawing?

Answer: _____ cm

- 9.** A logo triangle has area 7 square centimeters. For a poster it is enlarged by scale factor 3 (every length becomes 3 times as long). Find the area of the enlarged logo.

Answer: _____ cm^2

- 10.** An open-top aquarium is 60 cm long, 30 cm wide, and 40 cm tall. Glass is needed for the bottom and the four sides — but *not* the top. How much glass is needed?

Answer: _____ cm^2

- 11.** On a meter grid, a house-shaped sign has corners at $(0,0)$, $(8,0)$, $(8,5)$, $(4,8)$, and $(0,5)$ — a rectangle with a triangular top. Find the sign's area.

Answer: _____ m^2

12. Challenge. An architect builds a 1 : 20 scale model of a room that is 4 m long and 3 m wide. (Remember: 1 m = 100 cm.)

- (a) Model floor length: _____ Model floor width: _____
- (b) Model floor area, in cm^2 : _____
- (c) Explain why the model's floor area is *not* the real floor area divided by 20.